

Negative Exponent Law*Question Bank - Additional 3.04*

Show your work. Circle your final answer.

For questions 1 through 6, rewrite each expression as an equivalent expression with positive exponents. Do not evaluate.

1. $\frac{2^{-8}}{2^{-3}}$

2. $-7 \cdot 2^{-4}$

3. $\frac{1}{10^{-3}}$

4. $(-4)^{-5}$

5. 9^{-6}

6. $\frac{1}{(-3)^{-7}}$

For questions 7 and 8, apply the negative exponent law, then the product or quotient of powers law, and evaluate.

7. Evaluate each expression.

a. $(-7)^{-6} \cdot (-7)^4$

b. $8^{-5} \cdot 8^5$

c. $5^6 \cdot 5^{-3}$

8. Evaluate each expression.

a. $(-6)^{-8} \div (-6)^{-5}$

b. $3^{-5} \div 3^2$

c. $5^0 \div 5^{-2}$

Opposite or reciprocal.

9. Four expressions are shown: -10 , 10^{-1} , $(-10)^{-1}$, -10^{-1} .
- a. Give the value of each one.

- b. Which of the four is the reciprocal of 10, and which is the opposite of 10?

Working backwards.

10. Find the value of n that makes each statement true.

a. $3^n = \frac{1}{81}$

b. $10^n = 0.00001$

c. $\frac{6^2}{6^n} = 6^5$

11. Which is the value of 3^{-4} ?

- ☐ A. $\frac{1}{81}$
- ☐ B. $\frac{1}{12}$
- ☐ C. -81
- ☐ D. -12

12. Select ALL of the expressions whose value is $\frac{1}{81}$.

- ☐ A. 3^{-3}
- ☐ B. 3^{-4}
- ☐ C. -3^4
- ☐ D. 9^{-2}
- ☐ E. $\frac{3^{-6}}{3^{-2}}$
- ☐ F. $\left(\frac{1}{3}\right)^4$